

Data Science Capstone Project

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Executive Summary

The data was collected from the SpaceX API and the SpaceX Wikipedia page. I created a new column called “**class**” to label whether a landing was successful or not. Then, I explored the data using SQL, visualizations, Folium maps, and dashboards.

Important columns were selected as features for the models. Categorical variables were converted into numerical form using **one-hot encoding**. The data was also **standardized**, and **GridSearchCV** was used to find the best parameters for the machine learning models. Finally, I compared and visualized the **accuracy scores** of all models.

Four machine learning models were built: **Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors**. All models showed similar performance, with an accuracy of about **83.33%**. However, they tended to predict too many successful landings. More data is needed to improve model performance and accuracy.

Introduction

-Background:

The commercial space industry is growing rapidly. SpaceX offers the most competitive launch price, about \$62 million per launch, compared to \$165 million from other providers. This lower cost is mainly due to their ability to recover and reuse the first stage of the rocket. Now, another company, Space Y, wants to compete with SpaceX.

-Problem:

Space Y has asked us to develop a machine learning model to predict whether the first stage recovery will be successful.



SpaceX Falcon 9 Rocket – The Verge

Methodology

OVERVIEW OF DATA COLLECTION, WRANGLING,
VISUALIZATION, DASHBOARD, AND MODEL METHODS

Methodology

- Data collection : combined data from SpaceX public API and SpaceX Wikipedia page
- Perform data wrangling: classifying true landings as successful and unsuccessful otherwise
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform predictive analysis using classification models : Tuned models using GridSearchCV
- Perform interactive visual analytics using Folium and Plotly Dash

Data collection and data wrangling methodology

Data Collection

Data Collection Process:

-The data was collected using two main methods:

1.Sending API requests to the SpaceX public API

2.Web scraping a data table from the SpaceX Wikipedia page

-The next slide presents a flowchart for the API data collection process, and the following slide shows the web scraping process.

-SpaceX API Data Columns:

FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, Latitude

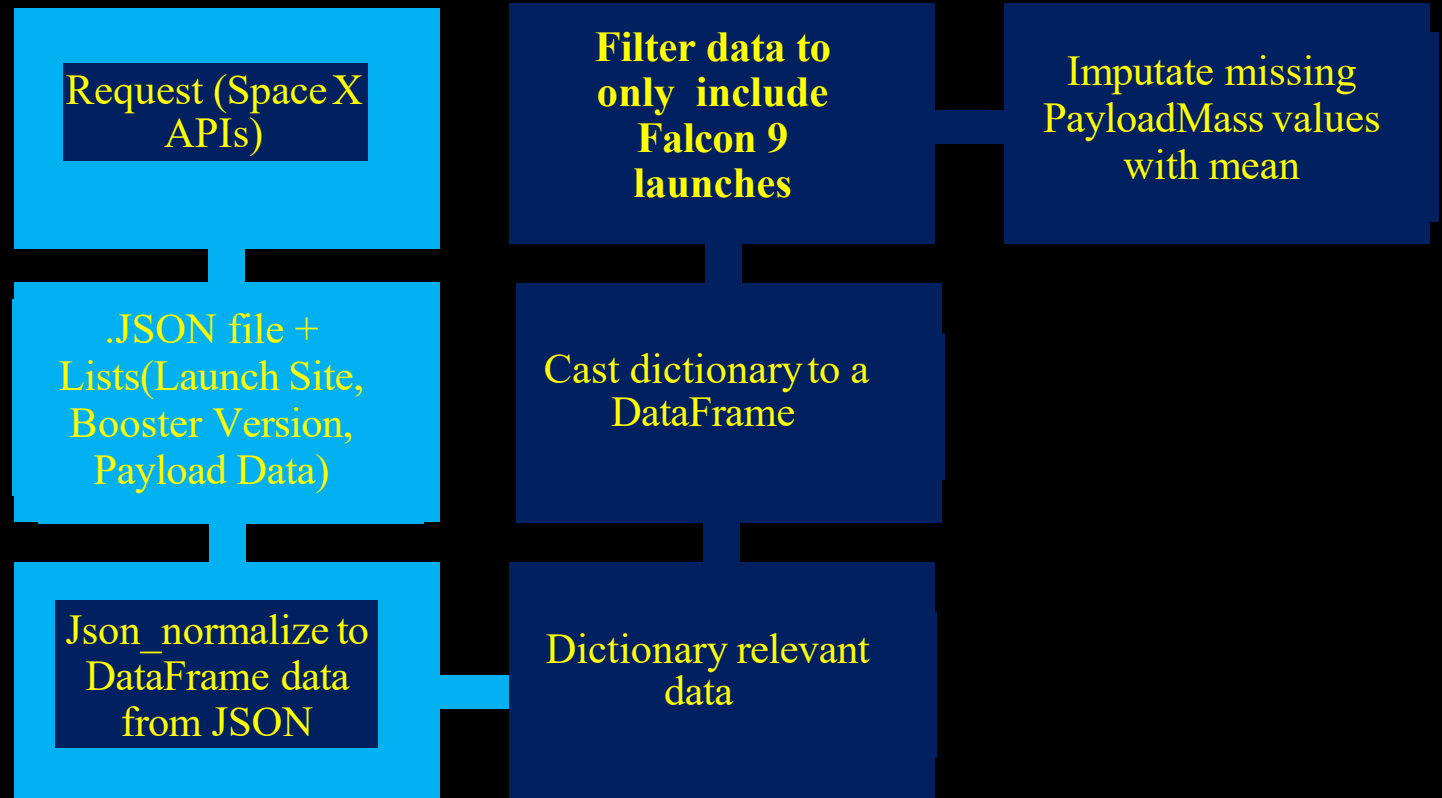
-Wikipedia Web Scraped Data Columns:

Flight No., Launch Site, Payload, PayloadMass, Orbit, Customer, Launch Outcome, Booster Version, Booster Landing, Date, Time

Data Collection — SpaceX API

GitHub url:

https://github.com/thayhiepth/Applied_Data_Science_Capstone/blob/main/1.Data%20Collection%20Api.ipynb



Data Collection— Web Scraping

GitHub url:

https://github.com/thayhiepth/Applied_Data_Science_Capstone/blob/main/2.Data%20Collection%20with%20Web%20Scraping.ipynb

Request Wikipedia
html

BeautifulSoup
html5lib Parser

Find launch
info html
table

Cast dictionary to
DataFrame

Iterate through
table cells to
extract data to
dictionary

Create dictionary

Data collection and data wrangling methodology

Data Wrangling

Creating the Training Label:

To prepare the data for modeling, we created a new training label based on the landing outcome. Successful landings were labeled as 1, and failed landings were labeled as 0.

The original Outcome column contains two parts:

- Mission Outcome
- Landing Location

A new column called “class” was created. It is assigned:

- 1 if the Mission Outcome is True (successful)
- 0 if the Mission Outcome is False or None (unsuccessful)

Value mapping:

- True ASDS, True RTLS, True Ocean → 1
- None None, False ASDS, None ASDS, False Ocean, False RTLS → 0
- GitHub url:

<https://github.com/thayhiepth/Applied Data Science Capstone/blob/main/3.Data%20wrangling.ipynb>

EDA and interactive visual analytics methodology

EDAwith Data Visualization

Exploratory Data Analysis (EDA):

Exploratory Data Analysis was performed on the following variables: **Flight Number, Payload Mass, Launch Site, Orbit, Class, and Year.**

Several plots were created to study the relationships between these variables, including:

- Flight Number vs. Payload Mass
- Flight Number vs. Launch Site
- Payload Mass vs. Launch Site
- Orbit vs. Success Rate
- Flight Number vs. Orbit
- Payload vs. Orbit
- Yearly success trend

We used **scatter plots, line charts, and bar charts** to compare the variables and identify patterns or relationships. These insights helped determine which features could be useful for training the machine learning models.

GitHub url:

https://github.com/thayhiepth/Applied_Data_Science_Capstone/blob/main/4.EDA%20with%20Visualization.ipynb

EDA and interactive visual analytics methodology

EDAwith SQL

SQL Analysis:

The dataset was loaded into an **IBM DB2 database** and queried using **SQL with Python integration**.

SQL queries were used to better understand the data and extract useful information. We analyzed details such as **launch site names, mission outcomes, customer payload sizes, booster versions, and landing results**.

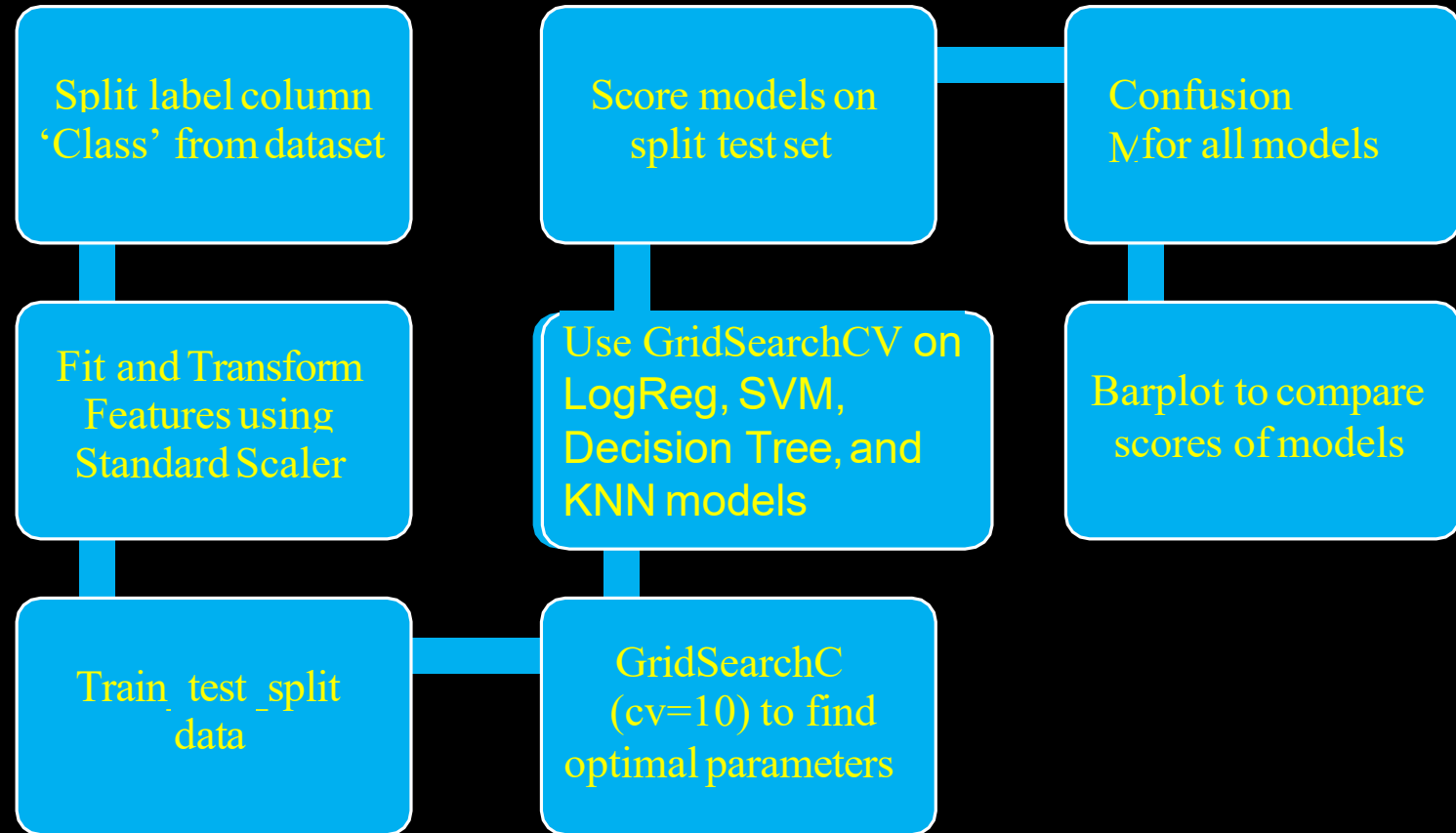
GitHub url:

https://github.com/thayhiepth/Applied_Data_Science_Capstone/blob/main/5.EDA%20with%20SQL.ipynb

Predictive analysis

GitHub url:

https://github.com/thayhiepth/Applied_Data_Science_Caps_tone/blob/main/7.Machine%20Learning%20Prediction.ipynb

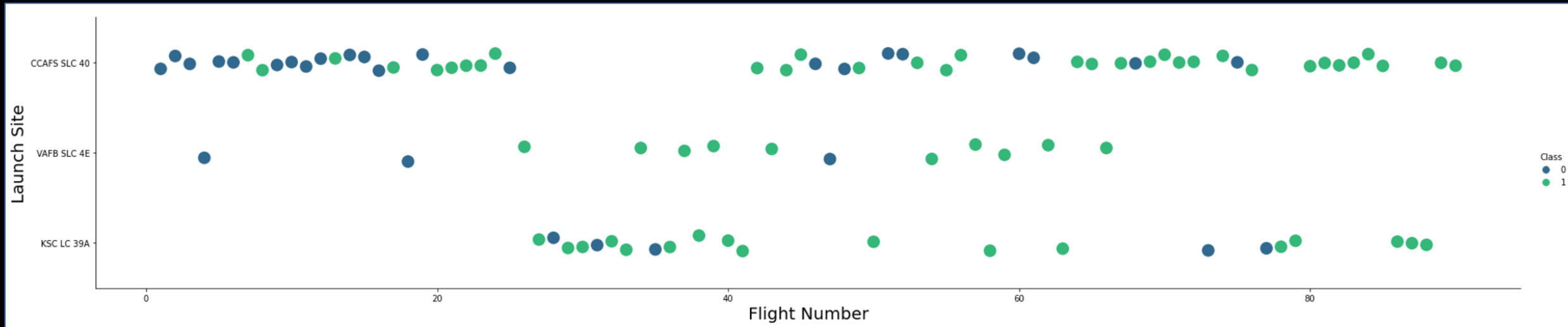


Results

EDA with visualization results slides

EDA with visualization results slides

Flight Number vs. LaunchSite

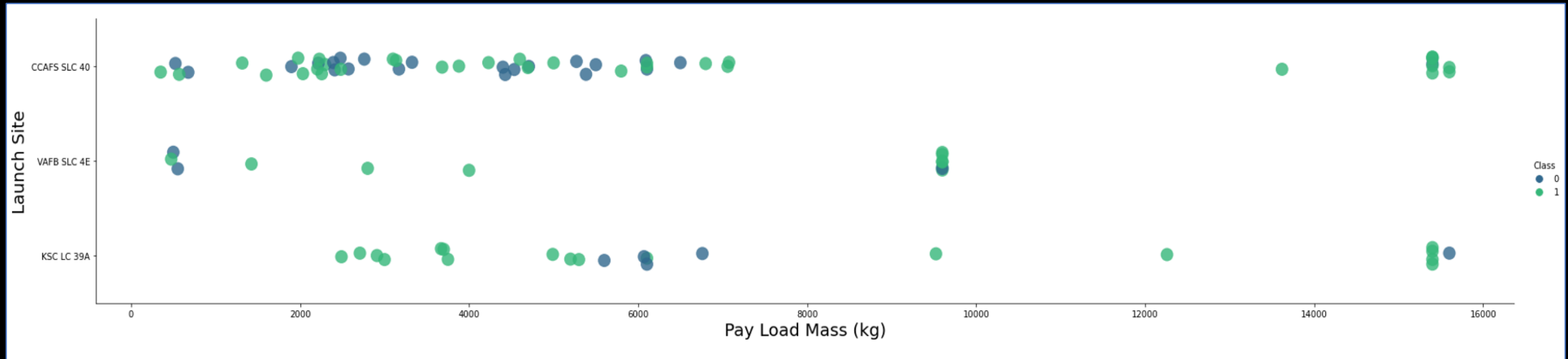


Green indicates successful launch; Purple indicates unsuccessful launch.

The graph shows that the success rate increases over time, as indicated by the flight number. There appears to be a major improvement around Flight 20, where the success rate rises significantly. The data also shows that CCAFS is the primary launch site, as it has the highest number of launches.

EDA with visualization results slides

Payload vs. Launch Site

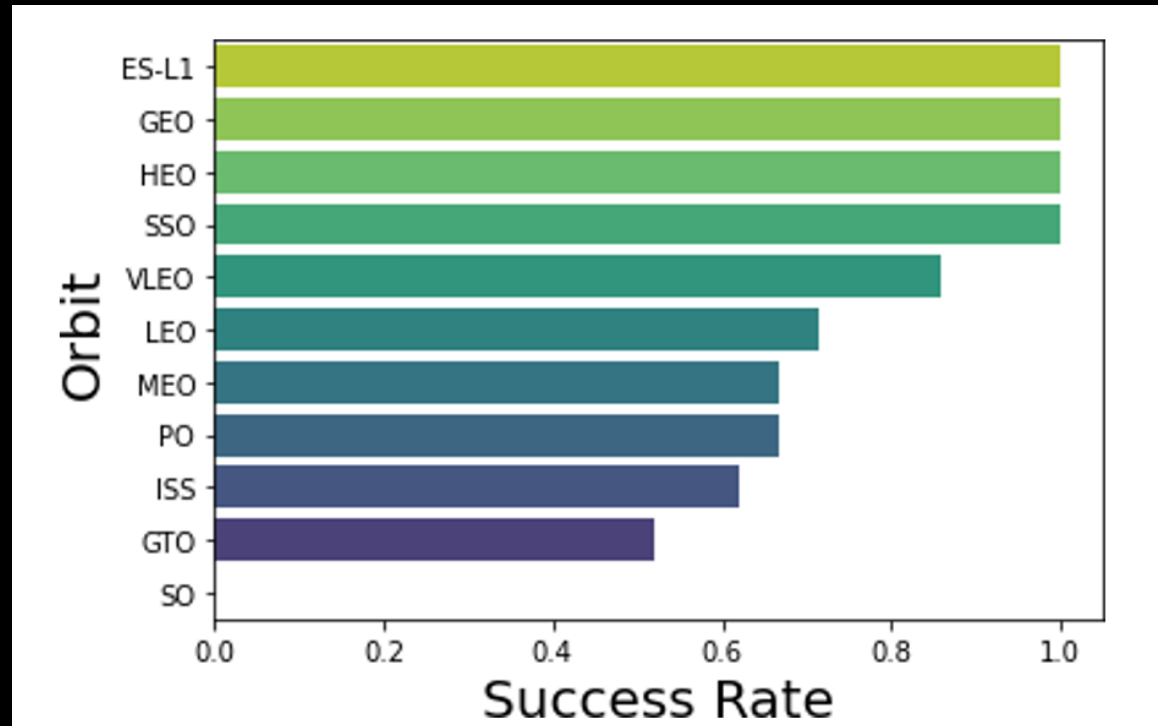


Green indicates successful launch; Purple indicates unsuccessful launch.

Most payload masses fall within the range of 0 to 6,000 kg. The data also suggests that different launch sites tend to handle different payload sizes, indicating variations in launch capabilities or mission types across sites.

EDA with visualization results slides

Success rate vs. Orbit type



Success Rate Scale
with 0 as 0%
0.6 as
60% 1
as 100%

Some orbits show very high success rates. **ES-L1, GEO, and HEO** each have a **100% success rate**, but their sample sizes are very small (1 launch each). **SSO** also has a **100% success rate** with 5 launches.

VLEO has a good number of attempts (14) and maintains a **fairly high success rate**.

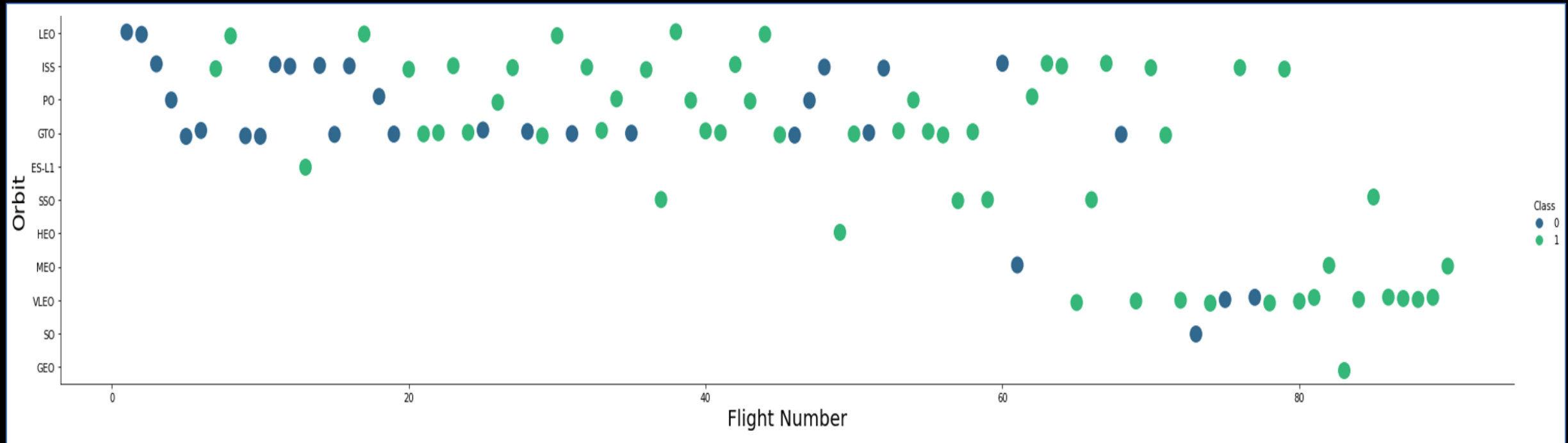
In contrast, **SO** has a **0% success rate**, but it only has 1 launch.

GTO has the **largest number of launches (27)**, with a success rate of around **50%**.

Overall, results with larger sample sizes are more reliable, while smaller samples may not accurately represent performance.

EDA with visualization results slides

Flight Number vs. Orbittype



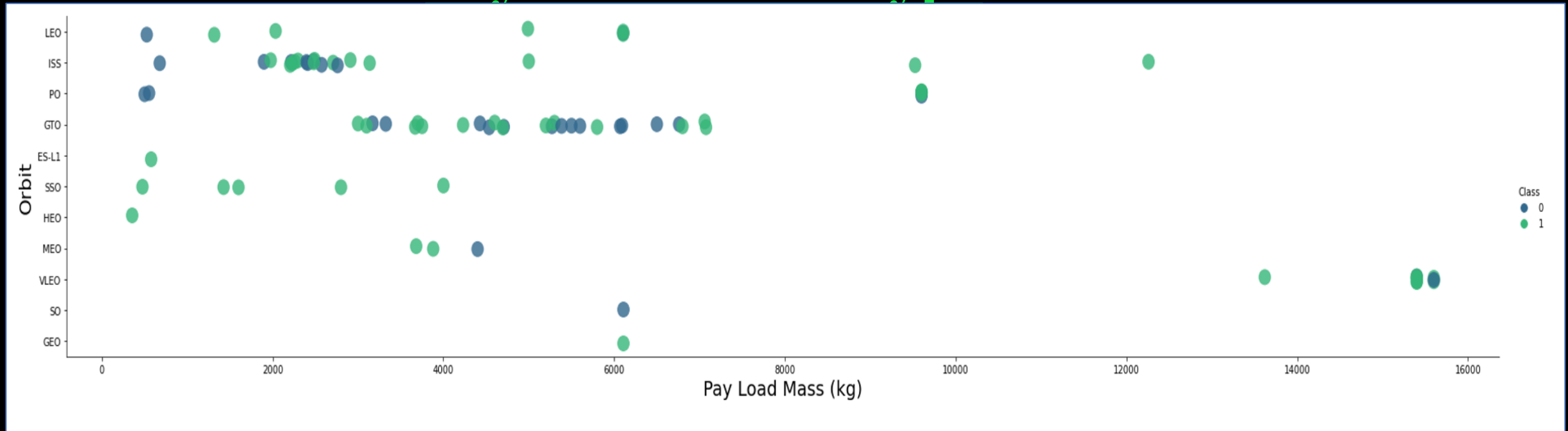
Green indicates successful launch; Purple indicates unsuccessful launch.

Launch orbit preferences changed over time, as shown by the flight number. The launch outcomes also appear to be related to these orbit choices.

At the beginning, SpaceX mainly used **LEO orbits**, which had moderate success. Later, the company returned to **VLEO orbits** in more recent launches. Overall, SpaceX seems to achieve **better success rates in lower orbits and Sun-synchronous orbits**.

EDA with visualization results slides

Payload vs. Orbit type



Green indicates successful launch; Purple indicates unsuccessful launch.

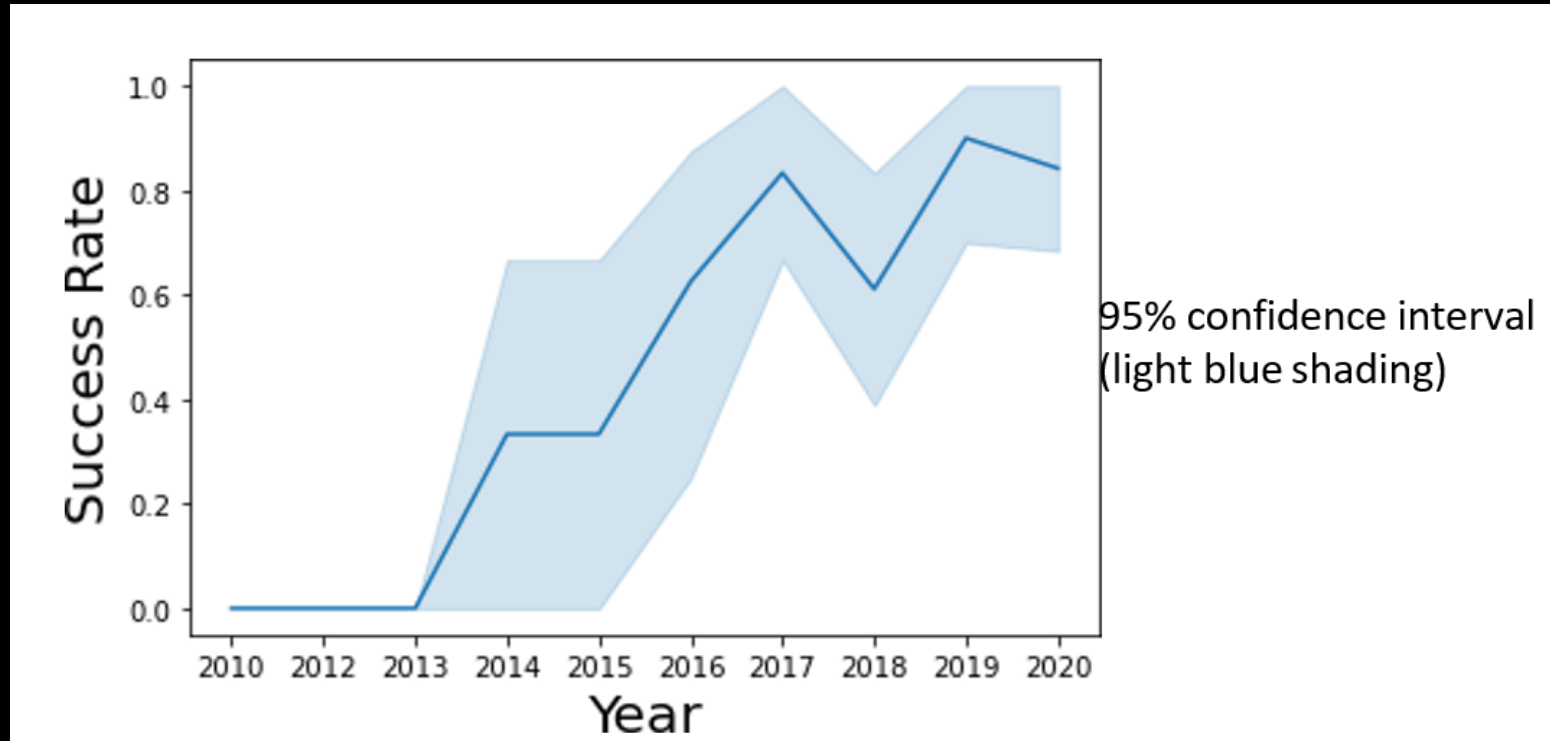
Payload mass appears to be related to the type of orbit.

LEO and SSO missions generally carry **lower payload masses**, while **VLEO**, another highly successful orbit, tends to carry **heavier payloads** at the higher end of the range.

This suggests that payload size may influence orbit selection and mission success.

EDA with visualization results slides

Launch Success Yearly Trend



The success rate has generally **increased over time since 2013**, with a small decline around **2018**. In recent years, the success rate has remained high at **around 80%**.

EDA with SQL results slides

EDA with SQL results slides

All Launch Site Names

In [4]:

```
%%sql
SELECT UNIQUE LAUNCH_SITE
FROM SPACEXDATASET;
```

```
* ibm_db_sa://ftb12020:***@0c77d6f2
Done.
```

Out[4]:

launch_site
CCAFS LC-40
CCAFS SLC-40
CCAFSSLC-40
KSC LC-39A
VAFB SLC-4E

We queried the database to find the **unique launch site names**.

The results showed that **CCAFS SLC-40** and **CCAFSSLC-40** likely refer to the same launch site, with differences caused by data entry errors. **CCAFS LC-40** appears to be an older name for the same location.

After cleaning the data, there are likely only **three unique launch sites**:

- CCAFS SLC-40
- KSC LC-39A
- VAFB SLC-4E

EDA with SQL results slides

Launch Site Names Beginning with CCA

In [5]:

```
%%sql
SELECT *
FROM SPACEXDATASET
WHERE LAUNCH_SITE LIKE 'CCA%'
LIMIT 5;
```

```
* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8lcg.databases.appdomain.cloud:31198/bludb
Done.
```

Out[5]:

DATE	time__utc__	booster_version	launch_site	payload	payload_mass__kg__	orbit	customer	mission_outcome	landing__outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

First five entries in database with Launch Site name beginning with CCA.

EDA with SQL results slides

Total Payload Mass from NASA

```
%%sql
SELECT SUM(PAYLOAD_MASS__KG_) AS SUM_PAYLOAD_MASS_KG
FROM SPACEXDATASET
WHERE CUSTOMER = 'NASA (CRS)';

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86
Done.
```

sum_payload_mass_kg

45596

This query calculates the **total payload mass (in kilograms)** for missions where **NASA** was the customer.

CRS stands for **Commercial Resupply Services**, which means these payloads were delivered to the **International Space Station (ISS)**.

EDA with SQL results slides

Average Payload Mass by F9v1.1

```
%%sql
```

```
SELECT AVG(PAYLOAD_MASS__KG_) AS AVG_PAYLOAD_MASS_KG  
FROM SPACEXDATASET  
WHERE booster_version = 'F9 v1.1'
```

```
* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86
```

```
Done.
```

avg_payload_mass_kg

2928

This query calculates the **average payload mass** for launches that used the **Falcon 9 v1.1 booster version**. The results show that the **average payload mass for F9 v1.1** is on the **lower end of the overall payload mass range**.

EDA with SQL results slides

First Successful Ground Pad Landing Date

```
%%sql
```

```
SELECT MIN(DATE) AS FIRST_SUCCESS  
FROM SPACEXDATASET  
WHERE landing__outcome = 'Success (ground pad)';
```

```
* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81  
Done.
```

first_success
2015-12-22

This query finds the **date of the first successful ground pad landing**. The first ground pad landing did not occur until **late 2015**. However, successful landings in general began to appear starting in **2014**.

EDA with SQL results slides

Successful Drone Ship Landing with Payload Between 4000 and 6000

```
%%sql
SELECT booster_version
FROM SPACEXDATASET
WHERE landing__outcome = 'Success (drone ship)' AND payload_mass__kg_ BETWEEN 4001 AND 5999;

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8lcg.database
Done.
```

booster_version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

This query returns the **four booster versions** that achieved **successful drone ship landings** with a **payload mass between 4,000 and 6,000 kg (exclusive)**.

EDA with SQL results slides

Total Number of Each Mission Outcome

```
%%sql
SELECT mission_outcome, COUNT(*) AS no_outcome
FROM SPACEXDATASET
GROUP BY mission_outcome;
```

```
* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-5
Done.
```

mission_outcome	no_outcome
Failure (in flight)	1
Success	99
Success (payload status unclear)	1

This query counts the number of each **mission outcome**. The results show that SpaceX achieves its **mission objectives about 99% of the time**, meaning most missions are successful. This suggests that many landing failures are **planned or expected**, rather than true mission failures. There is also **one launch with an unclear payload status** and **one launch that failed during flight**.

EDA with SQL results slides

Boosters that Carried Maximum Payload

```
%%sql
SELECT booster_version, PAYLOAD_MASS__KG_
FROM SPACEXDATASET
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXDATASET);

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1
Done.
```

booster_version	payload_mass__kg_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

This query returns the **booster versions** that carried the **highest payload mass of 15,600 kg**.

These boosters are very similar and all belong to the **F9 B5 (B10xx.x) series**. This suggests that **payload mass may be related to the booster version used**, with newer or upgraded boosters capable of carrying heavier payloads.

EDA with SQL results slides

2015 Failed Drone Ship Landing Records

```
%%sql
SELECT MONTHNAME(DATE) AS MONTH, landing__outcome, booster_version, PAYLOAD_MASS__KG_, launch_site
FROM SPACEXDATASET
WHERE landing__outcome = 'Failure (drone ship)' AND YEAR(DATE) = 2015;

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8lcg.databases.app
Done.
```

MONTH	landing__outcome	booster_version	payload_mass__kg_	launch_site
January	Failure (drone ship)	F9 v1.1 B1012	2395	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	1898	CCAFS LC-40

This query returns the **month**, **landing outcome**, **booster version**, **payload mass (kg)**, and **launch site** for launches in **2015** where **Stage 1** failed to land on a drone ship. The results show that there were **two such cases**.

EDA with SQL results slides

Ranking Counts of Successful Landings Between 2010-06-04 and 2017-03-20

```
%%sql
SELECT landing__outcome, COUNT(*) AS no_outcome
FROM SPACEXDATASET
WHERE landing__outcome LIKE 'Success%' AND DATE BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY landing__outcome
ORDER BY no_outcome DESC;

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8lce
Done.
```

landing__outcome	no_outcome
Success (drone ship)	5
Success (ground pad)	3

This query returns a list of **successful landings** between **June 4, 2010 and March 20, 2017 (inclusive)**. There are two types of successful landings: **drone ship landings** and **ground pad landings**. During this time period, there were **8 successful landings in total**.

Interactive map with Folium- related slides

Perform interactive visual analytics

Build an interactive map with Folium

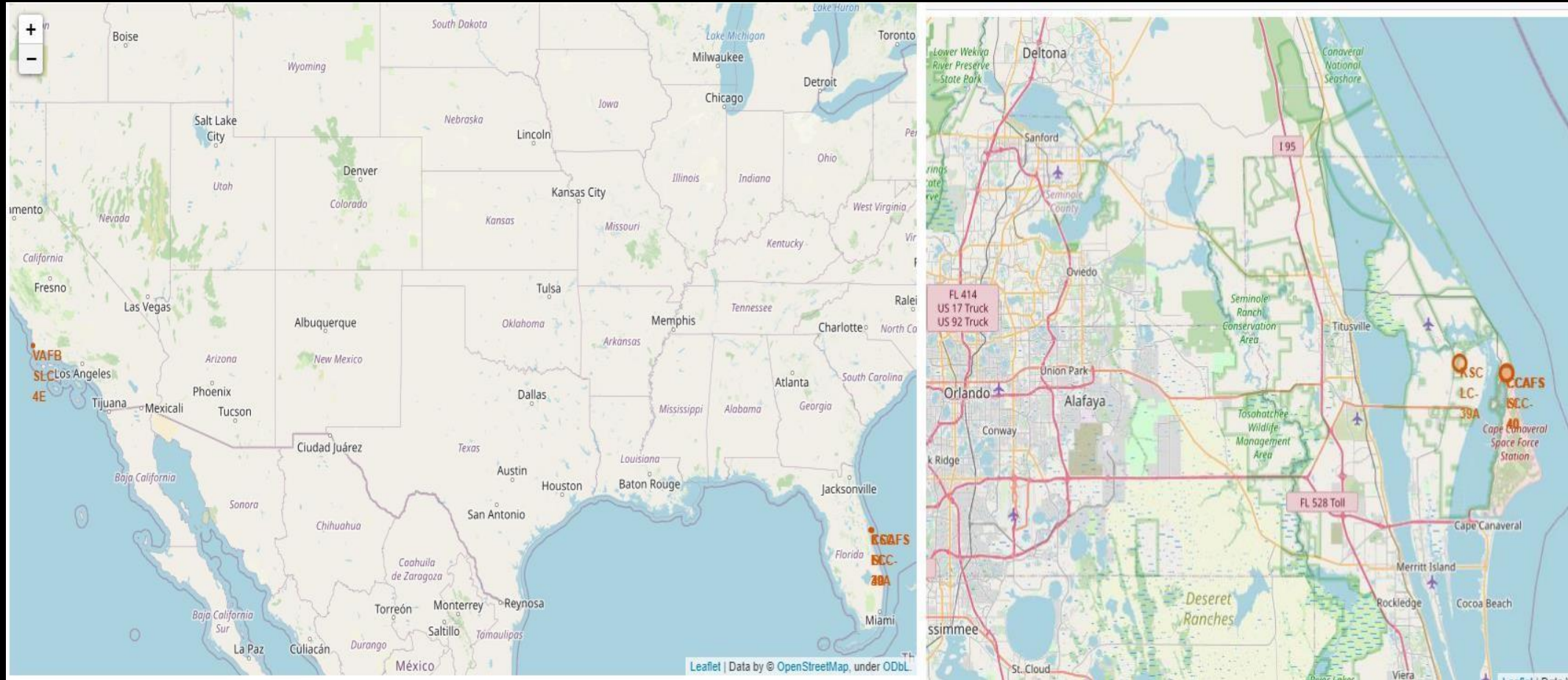
Folium Map Visualization: Folium maps were used to mark launch sites, as well as successful and unsuccessful landings. The maps also show the proximity of launch sites to key locations such as railways, highways, the coast, and nearby cities. This helps us understand why launch sites are placed in certain areas and allows us to visualize how landing success relates to location.

GitHub url:

https://github.com/thayhiepth/Applied_Data_Science_Capstone/blob/main/6.Interactive%20Visual%20Analytics%20with%20Folium.ipynb

Interactive map with Folium-related slides

Launch Site Locations

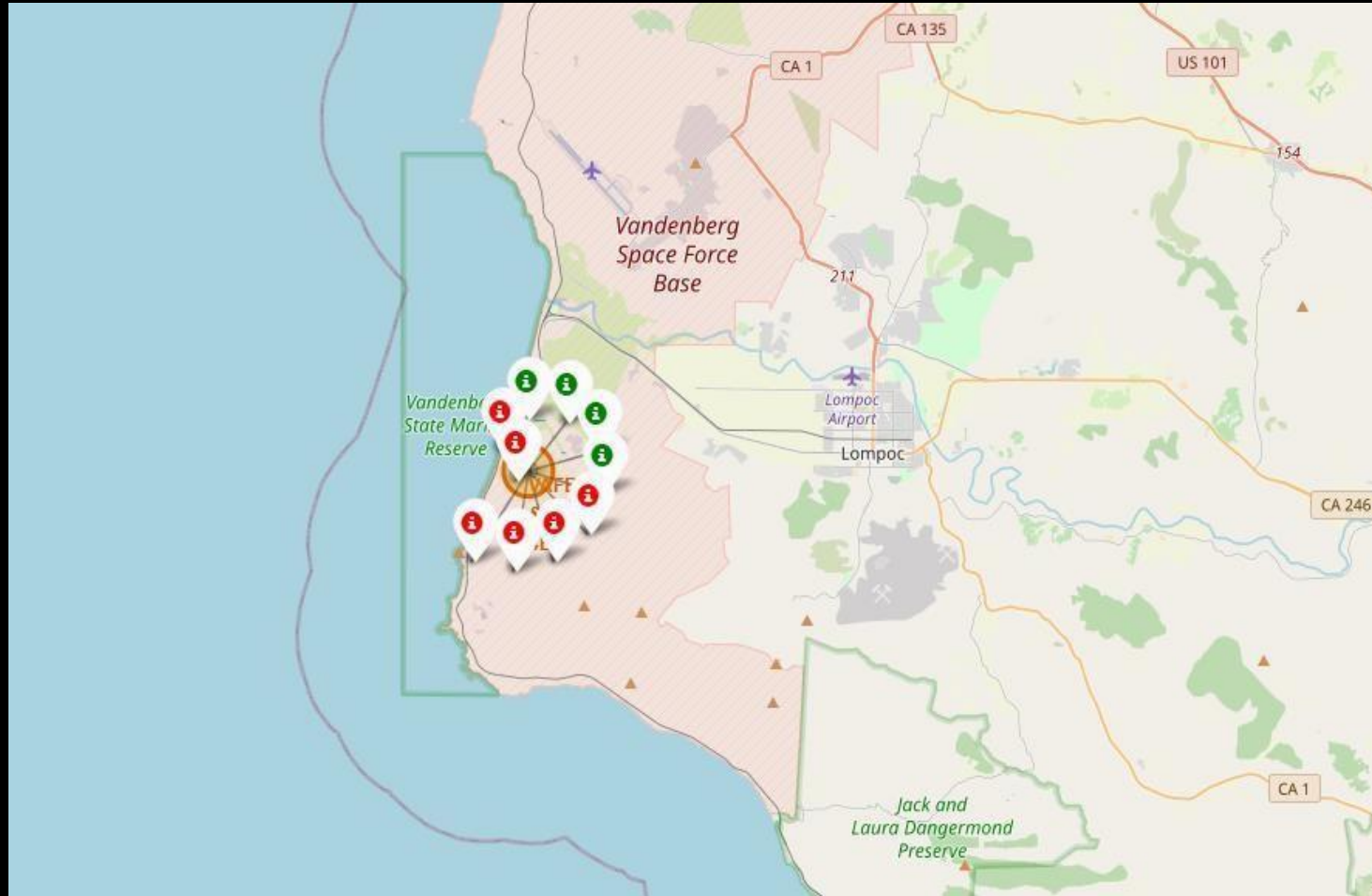


The **left map** shows all launch sites across the United States. The **right map** focuses on the two Florida launch sites, which are located very close to each other.

All launch sites are positioned **near the ocean**, likely for safety and easier rocket recovery operations.

Interactive map with Folium-related slides

Color-Coded Launch Markers

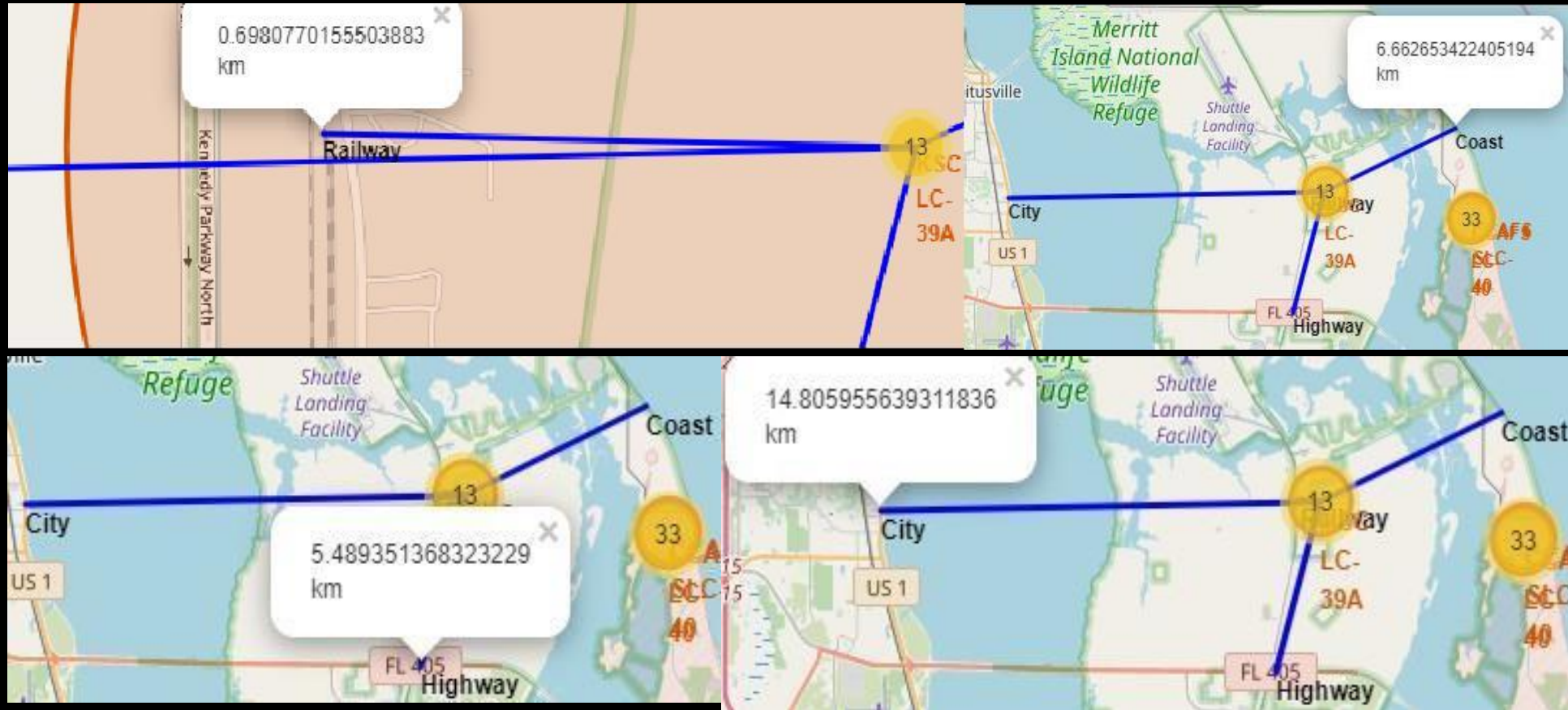


Clusters on the **Folium map** can be clicked to view individual landings. **Green icons** represent successful landings, and **red icons** represent failed landings.

In this example, **VAFB SLC-4E** shows **4 successful landings** and **6 failed landings**.

Interactive map with Folium-related slides

Key Location Proximities



Using **KSC LC-39A** as an example, the launch site is located close to **railways** to support the transport of large rocket parts and supplies. It is also near **highways** for easy movement of personnel and equipment. Additionally, the site is close to the **coast** and far from major cities. This location improves safety, as any launch failures are more likely to fall into the ocean rather than populated areas.

Plotly Dash- related slides

Perform interactive visual analytics

Build a Dashboard with PlotlyDash

Dashboard Visualization:

The dashboard includes a pie chart and a scatter plot to explore launch performance.

The pie chart shows the distribution of successful landings. Users can view results for all launch sites combined or select individual launch sites to see their specific success rates.

The scatter plot allows two inputs: selecting all sites or a single site, and adjusting the payload mass using a slider ranging from 0 to 10,000 kg.

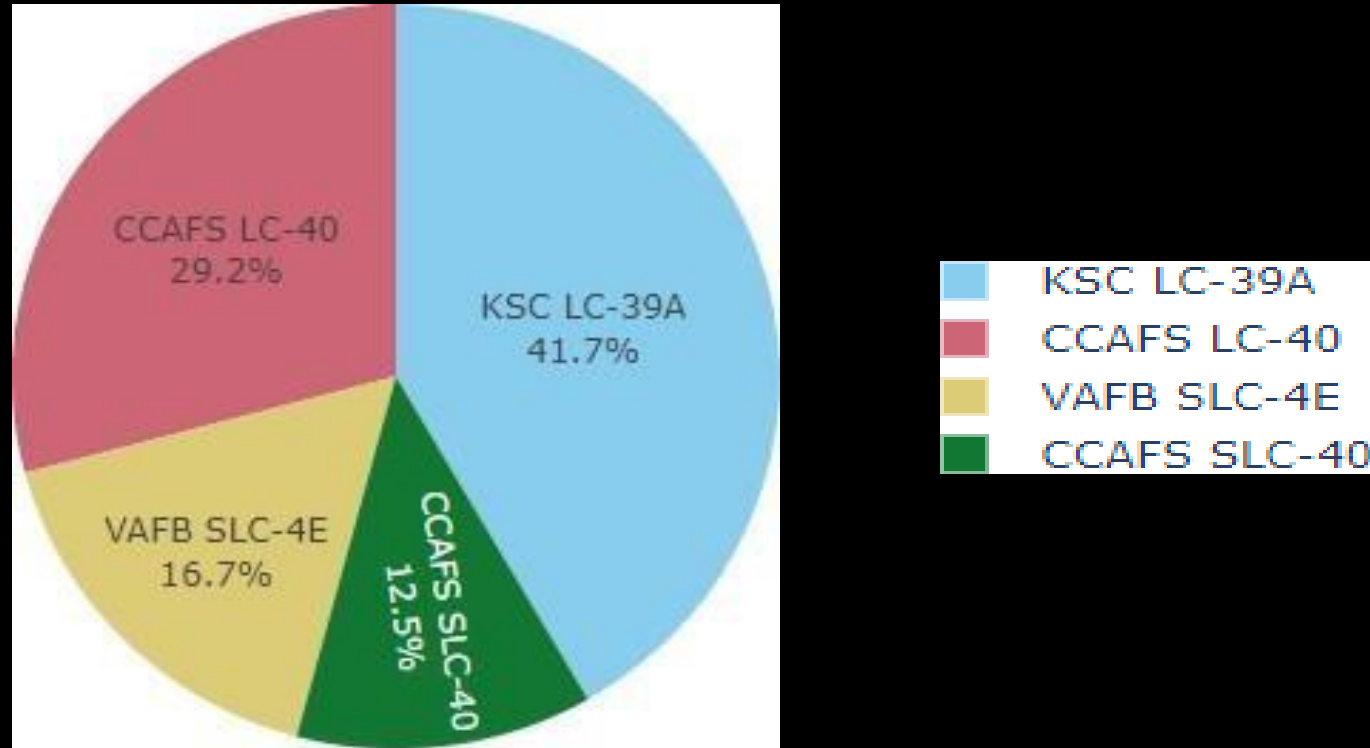
The pie chart helps visualize overall success rates by launch site, while the scatter plot shows how landing success changes based on launch site, payload mass, and booster version.

GitHub url:

https://github.com/thayhiepth/Applied_Data_Science_Capstone/blob/main/spacex_dash_app.py

Plotly Dash dashboard results slides

Successful Launches Across Launch Sites



This chart shows the **distribution of successful landings across all launch sites.**

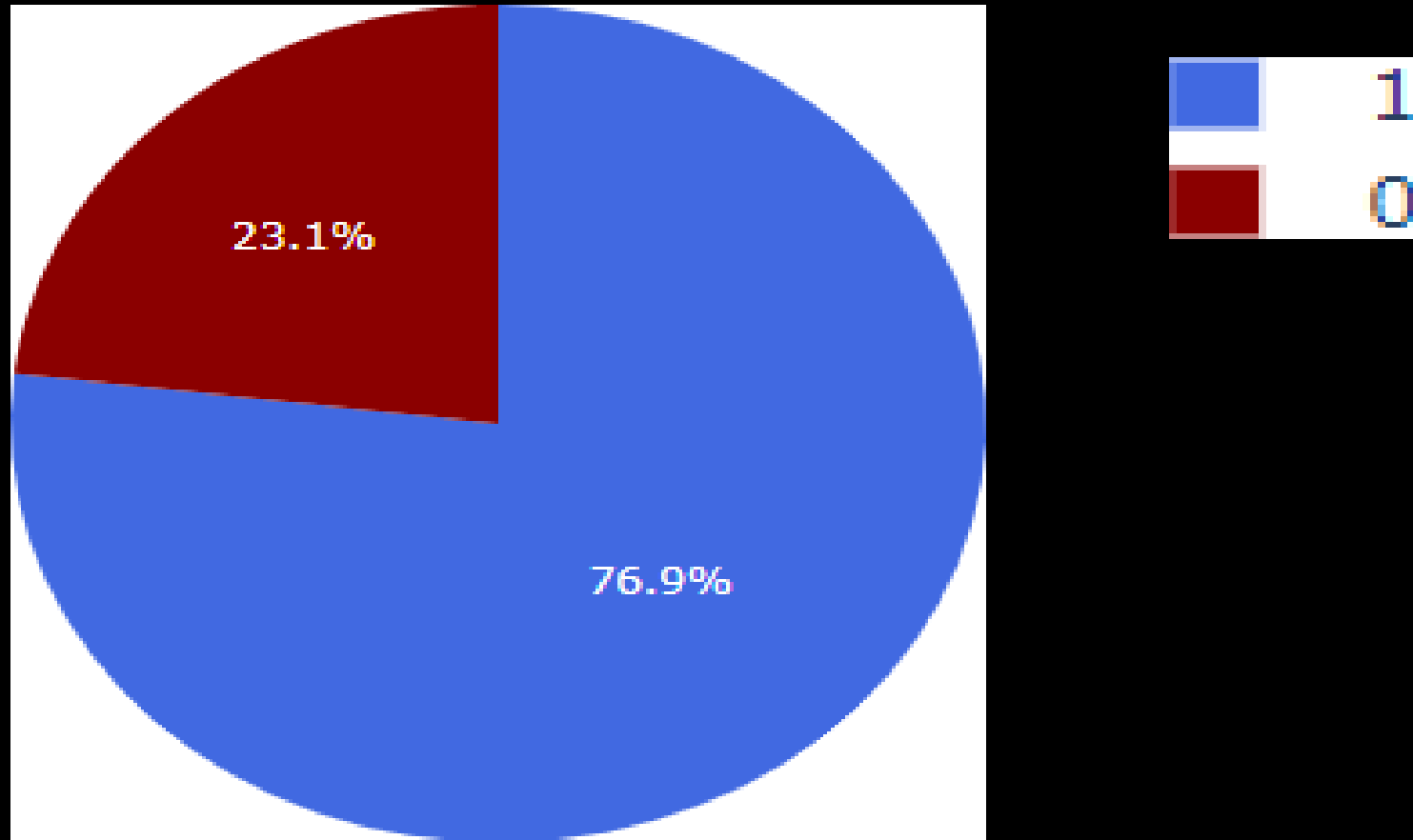
CAAFS LC-40 is the old name for **CAAFS SLC-40**, so when combined, **CAAFS and KSC have a similar number of successful landings.** Most of these successes occurred before the site name was changed.

VAFB has the **fewest successful landings**, which may be due to a smaller number of launches or the added difficulty of launching from the West Coast.

Plotly Dash dashboard results slides

Highest Success Rate Launch Site

KSC LC-39A Success Rate (blue=success)



KSC LC-39A has the highest success rate, with 10 successful landings and 3 failed landings.

Plotly Dash dashboard results slides

Payload Mass vs. Success vs. Booster Version Category



The Plotly dashboard includes a **payload range selector**, but it is currently set from **0 to 10,000 kg**, even though the maximum payload is **15,600 kg**.

In the scatter plot, **class = 1** represents a successful landing and **class = 0** represents a failure. The plot also uses **color to show booster version categories** and **point size to represent the number of launches**.

Within the **0–6,000 kg range**, there are two failed landings with **zero payload mass**, which is an interesting observation.

Predictive Analysis (classification) result slides

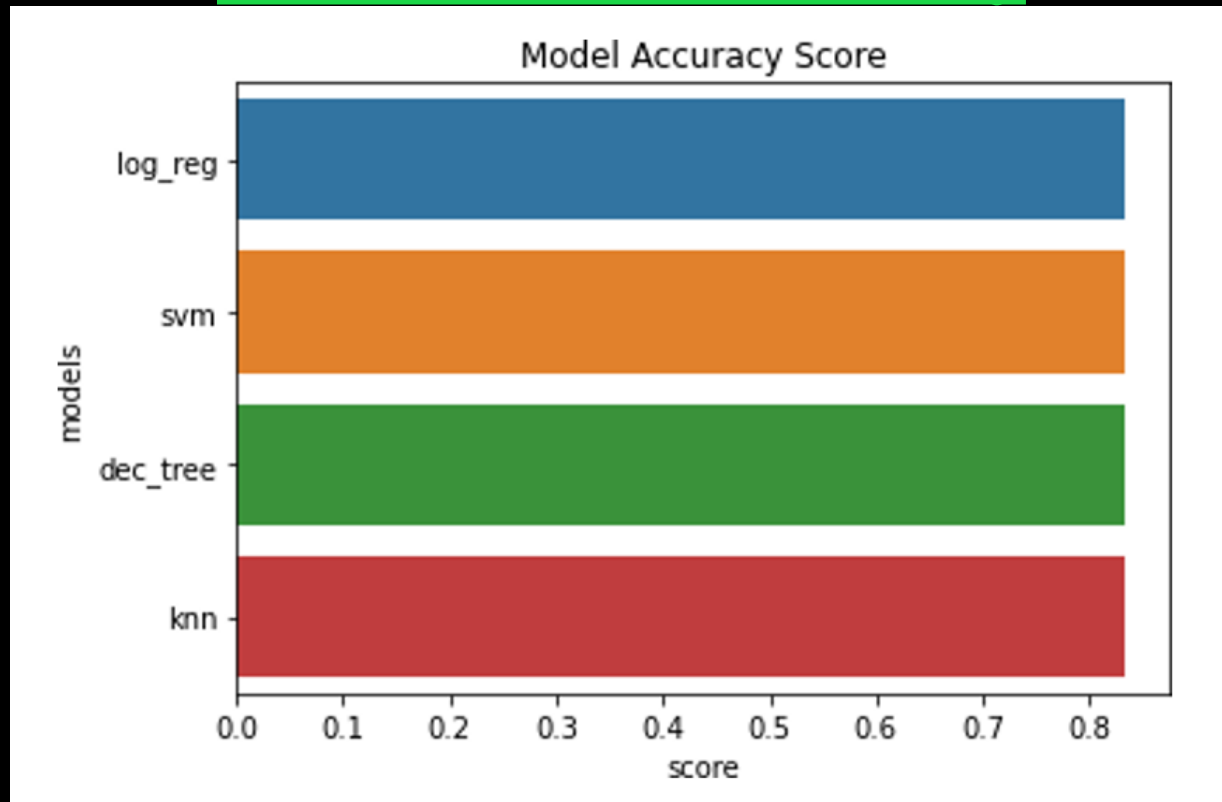
Predictive analysis (classification) results slides

PredictiveAnalysis

GRIDSEARCHCV(CV=10) ON LOGISTIC
REGRESSION, SVM, DECISION TREE,
AND KNN

Predictive analysis (classification) results slides

Classification Accuracy



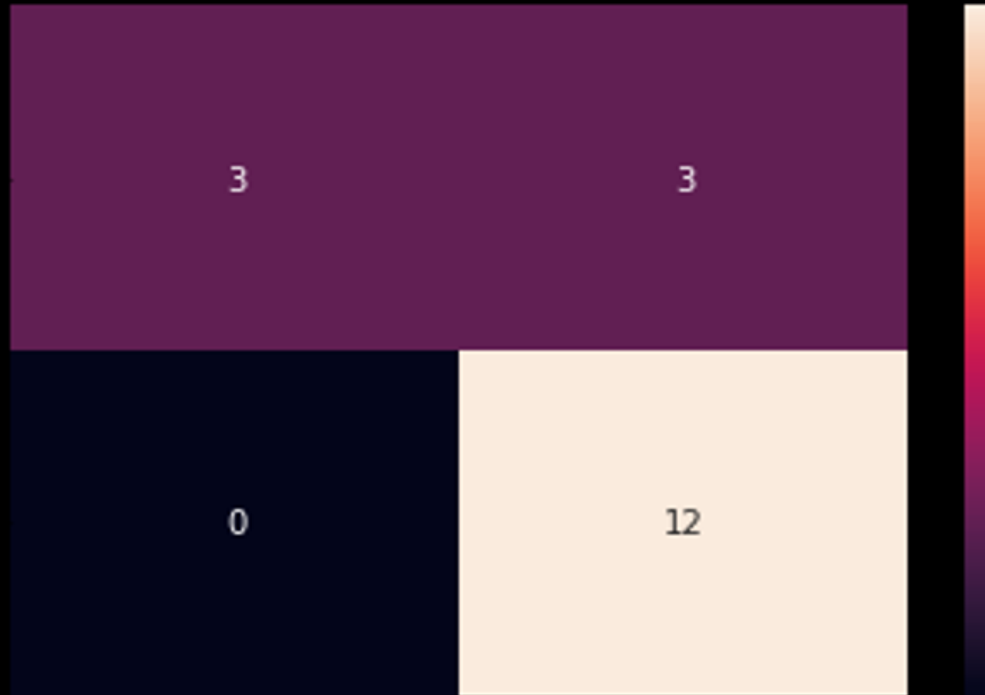
All models achieved nearly the same **test accuracy of 83.33%**. However, the **test set is small**, with only **18 samples**.

Because of this small size, the accuracy results may vary greatly between runs. For example, the **Decision Tree model** shows noticeable changes in performance when repeated.

To better evaluate and choose the best model, **more data is needed**.

Predictive analysis (classification) results slides

Confusion Matrix



Correct predictions are on a diagonal from top left to bottom right.

Since all models performed similarly on the test set, they produced the **same confusion matrix**.

The models correctly predicted:

- **12 successful landings** (true positives)
- **3 unsuccessful landings** (true negatives)

However, they incorrectly predicted:

- **3 successful landings when the landing actually failed** (false positives)

This shows that the models tend to **overpredict successful landings**.

CONCLUSION

Project Summary:

Our task was to develop a **machine learning model** for Space Y to help them compete with SpaceX. The goal of the model is to **predict whether Stage 1 will land successfully**, which could save about **\$100 million per launch** by allowing rocket recovery and reuse.

We collected data from the **SpaceX public API** and by **web scraping the SpaceX Wikipedia page**. The data was cleaned, labeled, and stored in a **DB2 SQL database**. We also built an **interactive dashboard** to visualize launch and landing trends.

Using this data, we trained several machine learning models and achieved an accuracy of about **83%**.

Space Y can use this model to estimate the likelihood of a successful Stage 1 landing before launch, helping them make better operational and financial decisions. However, collecting **more data** would likely improve the model's accuracy and help determine the best-performing approach.